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A Three-Tier Knowledge Graph RAG for Vietnamese Legal Question Answering

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Abstract. To answer legal questions accurately, a system must navigate hierarchical document structures, follow cross-references, and reason across multiple hops, but standard Retrieval Augmented Generation (RAG) systems cannot do this. Vietnamese legal text makes this gap worse. Tonal language, domain-specific abbreviations, and a rigid structural hierarchy (Law→Chapter→Article→Clause→Point) all compound retrieval difficulty. We propose a three-tier knowledge graph RAG architecture that extends the Legal-Onto Rela-model with a formal knowledge model $\mathscr{M} = (\mathscr{O}, \mathscr{G}, \mathscr{T}, \Phi)$, integrating an OWL ontology for terminological knowledge, a domain-specific knowledge graph for assertional knowledge, and a document tree forest for structural navigation. The three retrieval tiers, specifically LLM-guided tree traversal, dual-level semantic retrieval with intent-aware Personalized PageRank, and Reciprocal Rank Fusion, operate over these representations. We evaluate on 530 expert-annotated questions across three domains and 18 documents: enterprise law (197 questions), traffic safety law (203 questions), and postgraduate education regulation (130 questions). On legal questions, our system achieves 90.0% Hit@10 (MRR 0.596), outperforming PageIndex (84.8%), BM25 (47.5%), TF-IDF (45.8%), and LightRAG (39.8%). On education regulation, it reaches 88.5% Hit@10 (MRR 0.594), exceeding PageIndex by +13.9pp. Per-domain results confirm generalization: 83.2% on enterprise law, 96.6% on traffic law, and 88.5% on education, with no domain-specific tuning. We are not aware of prior work combining ontology-enhanced knowledge graphs with multi-tier retrieval for Vietnamese legal question answering.

Keywords. Knowledge Graph, Legal Question Answering, Retrieval Augmented Generation, Ontology, Vietnamese NLP, Multi-Tier Retrieval, Multi-Domain

1. Introduction

Legal forums and social networks in Vietnam are flooded with questions on business formation, traffic penalties, and regulatory compliance. Yet answers are often fragmented, contradictory, or cite repealed provisions.

Legal documents exhibit **complex hierarchical structures**: a single law organizes content into parts, chapters, sections, articles, clauses, and points, where each level carries distinct legal scope. They also contain pervasive **cross-references**: answering "If

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a member borrows money to meet charter capital and later withdraws it, what are the penalties?" requires chaining Article 47 (member obligations) to Article 16, Clause 5 (prohibited acts) across different chapters.

The **Vietnamese legal domain** adds further complexity. Vietnamese is tonal with ambiguous compound word boundaries. Legal practitioners use extensive abbreviations (*HDQT* for "Hoi dong quan tri", *CTCP* for "Cong ty co phan") that general models miss. Vietnamese law follows a rigid hierarchy such as Constitution, Codes, Decrees, and Circulars, where document type determines legal authority.

Traditional RAG systems [1] retrieve passages by vector similarity between query and document embeddings [2]. This works adequately for many domains [3,4], but legal QA exposes its weaknesses. Legal terminology carries precise meanings that general embeddings blur (**semantic gap**); documents organize content hierarchically, yet vector search treats them as flat text (**no structural awareness**); and answering a question often requires chaining references across articles, something dense retrieval cannot do (**no multi-hop reasoning**).

Existing approaches tackle parts of this problem. LightRAG [5] and GraphRAG [6] model entity relationships but disregard document hierarchy. PageIndex [7] exploits structure but has no semantic reasoning component. Legal ontology systems [8,9] formalize domain knowledge but remain disconnected from modern retrieval pipelines. We propose a **three-tier knowledge graph RAG** architecture that brings these capabilities together:

1. **LLM-guided tree traversal** through document hierarchies for structural navigation (Tier 1),
2. **Dual-level semantic retrieval** with intent-aware Personalized PageRank on a domain knowledge graph (Tier 2),
3. **Reciprocal Rank Fusion** for cross-tier merging with knowledge graph expansion (Tier 3).

Our contributions: (1) a three-tier RAG architecture combining structural, semantic, and graph-based retrieval; (2) a Vietnamese legal KG (5,936 entities, 5,633 relations) with semi-automatic construction; (3) a multi-intent PPR algorithm adapting edge weights to query intent. Experiments on 530 questions across three domains show 90.0% Hit@10 on legal QA (+5.2pp over PageIndex) and 88.5% on education regulation (+13.9pp over PageIndex), demonstrating generalization beyond legal text.

2. Related Work

RAG and KG-augmented retrieval. RAG [1] grounds generation in retrieved evidence. Since early dense retrieval [10], designs have added reflection [11], corrective retrieval [12], and hypothetical document generation [13]; surveys [3,4] trace this progression. GraphRAG [6] and LightRAG [5] introduce knowledge graphs for multi-hop reasoning but treat documents as bags of entities, losing structural context. Personalized PageRank [16,17] computes query-specific node importance. PageIndex [7] uses LLM-guided tree traversal for high precision but misses content outside the traversal path.

Legal NLP. Legal IR has moved from lexical methods [18,19] toward domain-adapted transformers such as LEGAL-BERT [20], which capture legal phrasing better than general models but still treat documents as flat passages. Ontology-based approaches [9] add richer structure; Legal-Onto [8] combines the Rela-model [21] with conceptual graphs for Vietnamese Land Law, though it relies on manual inference rules and was not designed for retrieval. For Vietnamese specifically, PhoBERT [22] and VnCoreNLP [23] provide foundational NLP capabilities, and Ngo et al. [24] combined BM25 with BERT [25] and domain heuristics, an effective baseline that nonetheless lacks graph-based reasoning. Surveys [26,27] confirm that bridging formal legal knowledge with scalable retrieval remains an open problem.

Positioning. Table 1 maps each system's capabilities. The gap motivating our work is the absence of a system that combines ontological knowledge representation with multi-tier retrieval for Vietnamese legal text.

Table 1. Feature comparison of legal QA approaches. *Not a RAG system; included for ontology comparison.

Category	Feature	NaiveRAG	LightRAG	GraphRAG	PageIndex	Legal-Onto*	Ours
Retrieval	Semantic Retrieval	✓	✓	–	–	–	✓
	Graph Traversal	–	✓	✓	–	–	✓
	Tree Navigation	–	–	–	✓	–	✓
Knowledge	Knowledge Graph	–	✓	✓	–	✓	✓
	Domain Ontology	–	–	–	–	✓	✓
Reasoning	Multi-hop	–	–	✓	–	✓	✓
	Intent-aware	–	–	–	–	–	✓
Integration	Multi-tier	–	–	–	–	–	✓
	Rank Fusion	–	–	–	–	–	✓
	Multi-domain	–	–	–	–	–	✓

3. Methodology

3.1. Problem Formulation

Given a legal corpus $\mathscr{D} = \{d_1, \ldots, d_m\}$ with hierarchically organized articles and a query q, retrieve a ranked list $A^* = [a_1^*, \ldots, a_k^*]$ of relevant articles. We define $A^* = \Phi(q, \mathscr{M})$ where $\mathscr{M} = (\mathscr{O}, \mathscr{G}, \mathscr{T}, \Phi)$ integrates ontology $\mathscr{O}$ (terminological), knowledge graph $\mathscr{G}$ (assertional), document forest $\mathscr{T}$ (structural), and retrieval functions Φ.

3.2. System Overview

Figure 1 illustrates the architecture. The **offline phase** constructs $\mathscr{M}$ through parsing, extraction, and ontology generation. The **online phase** processes queries via ontology expansion (Stage 0) followed by three retrieval tiers leveraging $\mathscr{T}$, $\mathscr{G}$, and $\mathscr{O}$.

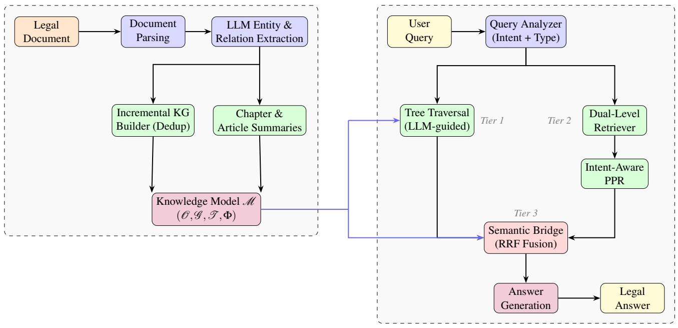

Figure 1. System architecture. Offline: knowledge model construction. Online: three-tier query processing (Tier 1: tree traversal, Tier 2: semantic+PPR, Tier 3: RRF fusion).

3.3. Knowledge Representation Model

Our model extends Legal-Onto [8], which defines $K = (C, R, RULES) + (Key, Rel, weight)$ integrating the Rela-model [21] with conceptual graphs. We extend this by separating terminological from assertional knowledge, adding structural representation, and replacing manual rules with LLM-based retrieval.

Definition 1 (Legal Knowledge Model)**.** *Given a legal corpus $\mathscr{D}$, the knowledge model is:*

$$\mathscr{M} = (\mathscr{O}, \mathscr{G}, \mathscr{T}, \Phi) \tag{1}$$

where $\mathscr{O}$ is a domain ontology, $\mathscr{G}$ is a knowledge graph, $\mathscr{T}$ is a document forest, and Φ is a set of retrieval functions.

Ontology $\mathscr{O} = (C, P, H, L)$ represents terminological knowledge: C = OWL classes for legal entity types, $P = P_{\text{obj}} \cup P_{\text{data}}$ = object and datatype properties, $H \subseteq C \times C$ = class hierarchy via `rdfs:subClassOf` (4-level taxonomy), and $L : C \cup P \to$ Vietnamese = label mapping (e.g., `JointStockCompany` $\mapsto$ "cong ty co phan"). This separates the TBox from Legal-Onto's flat concept set, enabling class hierarchy traversal for query expansion.

Knowledge Graph $\mathscr{G} = (E, R)$ represents assertional knowledge. Each entity $e = (id, name, type, conf, src)$ carries one of 14 types (Table 2), a confidence score, and source provenance. Each relation $r = (e_i, pred, e_j, evid, conf)$ spans 56 predicate types with evidence text; selected predicates are declared symmetric or transitive to support inference during PPR traversal.

In short, $\mathscr{O}$ answers "*what kind of entity?*" via taxonomy traversal (e.g., recognizing `JointStockCompany` and `LLC` as sibling subclasses of `Company`), while $\mathscr{G}$ answers "*what rules apply?*" by linking entities to governing articles through typed relations.

Document Forest $\mathscr{T} = \{T_1, \ldots, T_m\} \cup X$ represents structural knowledge. Each T_i is a rooted tree with six levels (Document $\to$ Chapter $\to$ Section $\to$ Article $\to$ Clause $\to$ Point), and X contains cross-reference edges between documents. The forest spans multiple legal domains, with documents organized into domain groups (e.g., enterprise law, traffic safety law) that enable domain-aware routing during retrieval.

Retrieval Functions $\Phi = \{\phi_0, \phi_1, \phi_2, \phi_3\}$ replace Legal-Onto's manual inference rules:

- $\phi_0 : Q \times \mathscr{O} \to Q'$ (ontology expansion),
- $\phi_1 : Q' \times \mathscr{T} \to 2^A$ (tree traversal),
- $\phi_2 : Q' \times \mathscr{G} \times \mathscr{O} \to 2^A$ (semantic retrieval),
- $\phi_3 : 2^A \times 2^A \times \mathscr{G} \to A^*$ (fusion).

Where Legal-Onto applies hand-crafted deductive rules, our functions leverage LLM reasoning while preserving formal relation properties in the graph structure.

3.4. Knowledge Graph Construction

The offline phase builds $\mathscr{G}$ in four steps: **(1) Document Parsing** into hierarchical trees forming $\mathscr{T}$; **(2) Entity-Relation Extraction** via one LLM call per article, producing evidence-grounded triples; **(3) Incremental KG Building** with Vietnamese slug normalization (NFD $\to$ diacritics removal $\to$ lowercasing, e.g., "Cong ty co phan" $\to$ *cong-ty-co-phan*) and a 4-pass deduplication pipeline across all 13 documents; **(4) Summary Generation** of chapter and article summaries for Tier 1 traversal. Following knowledge engineering practice in ontology-based expert systems [9], construction is **semi-automatic**: after LLM extraction, retrieval evaluation on a development query set revealed that LLM-generated relations missed most cross-document references. A domain expert then added $\sim$130 relations (cross-document and intra-document references, concept entities linking common legal terms to their governing articles). This evaluate-and-refine loop is standard in expert systems and contributes to the retrieval accuracy reported in Section 4.

3.4.1. Intent-Aware Edge Weights

Each query intent amplifies its most relevant relation types during PPR traversal. Specifically, each intent assigns weight 1.0 to its primary relation (*defines_as* for Definition, *requires* for Procedure, *has_authority* for Right, *has_obligation* for Obligation, *has_penalty* for Penalty) and 0.5 to all others. Two exceptions: *includes* receives 0.95 under Definition intent (near-synonym of *defines_as*), and *requires* receives 0.85 under Obligation intent (obligations often imply procedural requirements).

3.5. Three-Tier Retrieval

3.5.1. Stage 0: Query Analysis

The query analyzer performs: (1) **intent detection** across 6 types (Penalty, Definition, Procedure, Requirement, Reference, General) via Vietnamese regex with LLM fallback; (2) **type classification** into 7 categories; (3) **abbreviation expansion** (e.g., *TNHH* $\to$ *trach nhiem huu han*); and (4) **ontology expansion** $\phi_0(q, \mathscr{O})$ via class hierarchy traversal in H with Vietnamese label lookup through L.

3.5.2. Tier 1: LLM-Guided Tree Traversal

Tier 1 navigates $\mathscr{T}$ using LLM reasoning in three loops: **Loop 0** (Forest$\to$Domain$\to$Document) first matches the query to a domain group using pre-computed domain keywords, then selects top 5 documents within the matched domain; **Loop 1** (Document$\to$Chapter) selects top 8 chapters using summaries and keywords; **Loop 2** (Chapter→Article) identifies top 7 articles from article summaries. General Provisions chapters are automatically included. Each loop produces a reasoning path for interpretable retrieval.

3.5.3. Tier 2: Dual-Level Semantic Retrieval

Tier 2 performs hybrid semantic search over $\mathscr{G}$ and $\mathscr{O}$ with six components. *Low-level* (entity-specific): (1) keyphrase matching, (2) query-entity cosine similarity, (3) intent-aware PPR, (4) ontology class matching. *High-level* (theme/concept): (5) legal theme identification, (6) hierarchy traversal through $\mathscr{O}$'s taxonomy. The combined score is:

$$S_{\mathrm{T2}} = .05S_{\mathrm{kp}} + .20S_{\mathrm{con}} + .25S_{\mathrm{PPR}} + .20S_{\mathrm{sem}} + .15S_{\mathrm{thm}} + .15S_{\mathrm{hier}} \tag{2}$$

Multi-Intent Personalized PageRank. The PPR component computes query-specific node importance on $\mathscr{G}$:

$$\mathrm{PPR}(v) = (1-\alpha)\sum_{u\in N(v)} \frac{w(u,v)\cdot \mathrm{PPR}(u)}{d(u)} + \alpha \cdot p(v) \tag{3}$$

where $\alpha = 0.15$ is the teleport probability, $w(u,v)$ is the edge weight adjusted by query intent, $d(u)$ is the weighted out-degree of u, and $p(v)$ is the personalization vector seeded from matched entities.

3.5.4. Tier 3: Semantic Bridge (RRF Fusion)

Tier 3 merges results from Tier 1 and Tier 2 using Reciprocal Rank Fusion [28]:

$$\mathrm{RRF}(d) = \sum_{s\in\{T1,T2,KG\}} w_s \cdot \frac{1}{k + \mathrm{rank}_s(d)} \tag{4}$$

where $k = 60$ and source weights are $w_{T1} = 1.2$, $w_{T2} = 0.7$, $w_{KG} = 1.0$. Tree traversal receives the highest weight as it provides the most reliable structural signal; the moderate gap between w_{T1} and w_{KG} allows DualLevel and KG expansion to compensate when the LLM selects suboptimal chapters. KG expansion supplements with cross-reference articles outside the traversal path.

Knowledge graph expansion. After initial fusion, we expand results through $\mathscr{G}$'s relation edges. Strong relations (*references*, *requires*, *depends_on*) are prioritized for cross-chapter linking (up to 5 articles), while expansion relations (*defined_as*, *related_to*) handle same-chapter augmentation (up to 2 articles). Cross-document references are explicitly resolved via document-prefixed entity IDs.

Adjacent article expansion. After RRF scoring, we expand ±2 articles from each result with decay factor 0.7, as correct answers often lie in neighboring articles. Top-ranked articles are passed to an LLM for answer generation.

4. Experiments

4.1. Experimental Setup

Dataset. 18 Vietnamese documents across three domains: (1) *Enterprise law*: Law 59/2020/QH14 + 9 decrees, 77 chapters, 840 articles; (2) *Traffic safety law*: Law 36/2024/QH15 +

2 penalty decrees; (3) *Education regulation*: 5 postgraduate regulation documents from UIT/VNUHCM, 109 articles covering enrollment, degree requirements, and thesis procedures. Total: 949 articles. Evaluation: 530 expert-annotated questions (197 enterprise, 203 traffic, 130 education; 1–8 relevant articles each, avg. 2.3) from legal platforms LuatVietnam and ThuVienPhapLuat (legal) and university regulation portals (education).

Knowledge graph statistics. Table 2 summarizes the constructed knowledge graph.

Table 2. Knowledge graph statistics.

Entity Type	Count	Relation Type	Count	Structure	Count
Total Entities	5,936	Total Relations	5,633	Tree Nodes	9,226
Action	1,903	requires	1,377	Documents	13
Legal Term	1,632	applies_to	810	Chapters	77
Organization	500	has_authority	506	Articles	840
Legal Reference	451	references	434	Cross-refs	168
Other (9 types)	1,450	Other (52 types)	2,506		

Implementation. Claude 3.5 Haiku for offline entity extraction and knowledge graph construction. Claude Sonnet 4 for online tree traversal, query analysis, and evaluation. Embeddings: `paraphrase-multilingual-MiniLM-L12-v2` [2]. PPR: iterative computation, max 50 iterations, tolerance 10^{-6}. All LLM responses cached in SQLite for reproducibility.

Metrics. Hit@k, Recall@k, Precision@k, NDCG@k at $k \in \{1, 3, 5, 10, 20, 30\}$, and MRR.

4.2. Baselines

BM25 [18] ($k_1 = 1.2$, $b = 0.75$); **TF-IDF** [19] with Vietnamese segmentation; **Semantic Search** [2] using multilingual sentence-transformers; **LightRAG** [5] with Vietnamese entity extraction and default parameters, using the same LLM backend as our system; **PageIndex** [7] with two-step hierarchical tree search (document→chapter selection, then chapter→article selection), using heuristic summaries extracted from article text. All baselines use Sonnet 4 for online queries and retrieve from the full corpus with max_results=30.

4.3. Main Results

Table 3 presents Hit@k performance on the legal domain (400 questions).

Table 3. Legal domain Hit@k (%) on 400 questions. Best **bold**, 2nd <u>underlined</u>.

Method	Hit@5	Hit@10	Hit@20	Hit@30	MRR
Traditional retrieval					
Semantic	22.8	32.8	51.8	60.5	0.146
LightRAG [5]	25.5	39.8	47.8	49.2	0.173
BM25	30.8	47.5	66.2	74.8	0.200
TF-IDF	29.2	45.8	64.5	74.8	0.207
Keyword	33.5	48.5	63.2	72.8	0.222
RAG systems					
PageIndex [7]	<u>76.2</u>	<u>84.8</u>	<u>90.2</u>	<u>91.0</u>	<u>0.559</u>
3-Tier (ours)	**72.8**	**90.0**	**92.0**	**93.0**	**0.596**

Our 3-Tier system achieves 90.0% Hit@10 (MRR 0.596), outperforming PageIndex by +5.2pp and the best traditional baseline Keyword by +41.5pp. Among RAG systems, PageIndex (84.8%) benefits from tree-based structural navigation but misses cross-reference articles outside the traversal path. LightRAG (39.8%) uses Vietnamese entity extraction (corrected from English-language extraction in our initial evaluation); despite this fix, it underperforms even BM25 (47.5%) and keyword matching (48.5%) because its generic knowledge graph entities cause noise in hybrid round-robin merging, pushing correct vector matches out of the top 10. Traditional methods (BM25/TF-IDF/Keyword, ∼46–49%) lack structural awareness; semantic search (32.8%) struggles with Vietnamese legal terminology.

4.4. Multi-Domain Performance

Table 4 compares per-domain Hit@10 across all methods.

Table 4. Per-domain Hit@10 (%) across all methods and three domains.

Method	Enterprise (197)	Traffic (203)	Education (130)
Semantic	29.9	35.5	50.0
LightRAG	20.3	58.6	53.1
BM25	31.5	63.1	70.0
TF-IDF	39.1	52.2	68.5
Keyword	20.3	75.9	56.2
PageIndex	<u>76.6</u>	<u>92.6</u>	<u>74.6</u>
3-Tier (ours)	**83.2**	**96.6**	**88.5**
Ours vs. PageIndex	*+6.6pp*	*+4.0pp*	*+13.9pp*

Our system outperforms all baselines across all three domains. Traffic law achieves the highest Hit@10 (96.6%) because penalty articles map directly to specific violations.

Enterprise law is harder (83.2%) due to cross-decree references spanning multiple documents. Education regulation (88.5%) demonstrates that the architecture generalizes to non-legal regulatory text: postgraduate enrollment rules, degree requirements, and thesis procedures follow a similar hierarchical structure. The largest margin over PageIndex occurs in education (+13.9pp), where our dual-level semantic retrieval captures domain-specific terminology that tree navigation alone misses. No domain-specific tuning was applied; identical pipeline, parameters, and RRF weights serve all three domains.

4.5. Cross-Domain Generalization: Education

Table **??** presents full results on the education domain (130 questions, 5 documents), demonstrating generalization beyond legal text.

Table 5. Education domain Hit@k (%) on 130 questions.

Method	**Hit@5**	**Hit@10**	**Hit@20**	**Hit@30**	**MRR**
LightRAG [5]	24.6	53.1	72.3	77.7	0.197
Semantic	30.0	50.0	70.8	77.7	0.214
Keyword	36.9	56.2	75.4	85.4	0.242
TF-IDF	54.6	68.5	85.4	88.5	0.338
BM25	52.3	70.0	80.0	86.2	0.355
PageIndex [7]	62.3	74.6	79.2	80.0	0.441
3-Tier (ours)	**78.5**	**88.5**	**93.1**	**94.6**	**0.594**

Our system outperforms PageIndex by +13.9pp Hit@10 in this domain. Traditional baselines perform comparatively better here (BM25 reaches 70.0%) because education documents are smaller and terminology is less ambiguous than legal text. Nevertheless, structural navigation remains essential: PageIndex (74.6%) outperforms all traditional methods, and our three-tier approach adds another +13.9pp through KG-based cross-article linking.

4.6. Analysis

4.6.1. Performance by Query Category

On the legal domain, Hit@10 ranges from 0% (household business, 4 queries) to 98% (traffic rules and penalties) across categories. Dissolution (96.0%), traffic violations (97.6%), and speed-related penalties (100%) score highest, as these map cleanly to localized article clusters. Company establishment (57.9%) and company types (64.7%) score lowest because answers span multiple decrees outside the primary enterprise law.

4.6.2. Qualitative and Error Analysis

Two examples illustrate how the tiers cooperate. (1) *Multi-hop*: “If a member borrows money to meet charter capital and later withdraws it, what are the penalties?”. Tier 1 reaches Art. 47 (member obligations); PPR then walks *requires* edges from “charter capital” to Art. 16 Cl. 5 (prohibited acts); RRF places both in top 5. (2) *Cross-chapter*: “A partner contributed a house worth 1 bn VND, now worth 1.5 bn, can they withdraw it?”.

Tier 1 reaches Art. 178–180 (partnership rules); PPR adds Art. 186 (capital transfer) via *condition_for* edges. Neither example is answered by any baseline at Hit@30.

Of the 40 missed queries (10.0%), 33 are enterprise and only 7 are traffic. The primary failure modes: 8 stem from company establishment queries requiring articles across multiple decrees (42% miss rate in this category), 6 involve company type distinctions that span different legal documents, and 4 concern household business registration (100% miss rate, as governing articles lie in decrees outside our primary corpus). An additional 8 queries are borderline misses that appear at Hit@20 but not Hit@10. Queries with more than 2 expected articles account for 17 of the 40 misses, confirming that multi-article retrieval remains the hardest sub-problem.

5. Conclusion

This paper introduced a three-tier knowledge graph RAG system for Vietnamese legal QA, built on a formal knowledge model $\mathscr{M} = (\mathscr{O}, \mathscr{G}, \mathscr{T}, \Phi)$ that extends Legal-Onto by separating terminological, assertional, and structural representations. On 530 questions spanning three domains, the system achieves 90.0% Hit@10 (MRR 0.596) on legal questions, outperforming PageIndex (84.8%) by +5.2pp, and 88.5% Hit@10 on education regulation, exceeding PageIndex by +13.9pp.

The central finding is that structural navigation and semantic graph retrieval are complementary: Tier 1's tree traversal provides the most reliable structural signal (hence $w_{T1} = 1.2$), while Tier 2's intent-aware PPR catches cross-reference articles that lie outside the traversal path. This complementarity holds across domains: 83.2% Hit@10 on enterprise law, 96.6% on traffic safety law, and 88.5% on education regulation, despite substantial structural differences and no domain-specific tuning. Adding a new domain requires only parsing documents and running the same KG construction pipeline, as demonstrated by the education domain experiment.

Limitations. Tier 1 requires multiple LLM calls ($\sim$7 s latency per query). Entity extraction lacks gold-standard validation. Enterprise law remains the hardest domain (83.2%), with multi-article queries accounting for most failures.

Future work. We plan to extend coverage to additional regulatory domains (healthcare, finance), add multi-turn conversational support, and produce explainable retrieval traces citing graph paths used in ranking. The education domain results suggest the architecture generalizes well to hierarchically structured regulatory documents beyond legal text.

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References

[1] Lewis P, Perez E, Piktus A, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In: Advances in Neural Information Processing Systems 33 (NeurIPS 2020); 2020. p. 9459-74.

[2] Reimers N, Gurevych I. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP); 2019 Nov 3-7; Hong Kong. 2019. p. 3982-92, doi: 10.18653/v1/D19-1410.

[3] Gao Y, Xiong Y, Gao X, et al. Retrieval-Augmented Generation for Large Language Models: A Survey. arXiv preprint. 2024. arXiv:2312.10997, doi: 10.48550/arXiv.2312.10997.

[4] Fan W, Ding Y, Ning L, et al. A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2024); 2024. p. 6491-501, doi: 10.1145/3637528.3671470.

[5] Guo Z, et al. LightRAG: Simple and Fast Retrieval-Augmented Generation. arXiv preprint. 2024. arXiv:2410.05779, doi: 10.48550/arXiv.2410.05779.

[6] Edge D, Trinh H, Cheng N, et al. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. arXiv preprint. 2024. arXiv:2404.16130, doi: 10.48550/arXiv.2404.16130.

[7] Zhang M, Tang Y, et al. PageIndex: Next-Generation Vectorless, Reasoning-based RAG. 2025. Available from: https://pageindex.ai/blog/pageindex-intro.

[8] Nguyen T, Nguyen H, Pham V, Tran D, Selamat A. Legal-Onto: An Ontology-based Model for Representing the Knowledge of a Legal Document. In: Proceedings of the 17th International Conference on Evaluation of Novel Approaches to Software Engineering (ENASE 2022); 2022. p. 426-34, doi: 10.5220/0011088400003176.

[9] Sartor G, et al. Approaches to Legal Ontologies: Theories, Domains, Methodologies. Dordrecht: Springer; 2011. 272 p., doi: 10.1007/978-94-007-0120-5.

[10] Karpukhin V, Oguz B, Min S, et al. Dense Passage Retrieval for Open-Domain Question Answering. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP); 2020. p. 6769-81, doi: 10.18653/v1/2020.emnlp-main.550.

[11] Asai A, Wu Z, Wang Y, Sil A, Hajishirzi H. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. In: Proceedings of the International Conference on Learning Representations (ICLR 2024); 2024, doi: 10.48550/arXiv.2310.11511.

[12] Yan SQ, et al. Corrective Retrieval Augmented Generation. arXiv preprint. 2024. arXiv:2401.15884, doi: 10.48550/arXiv.2401.15884.

[13] Gao L, Ma X, Lin J, Callan J. Precise Zero-Shot Dense Retrieval without Relevance Labels. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023); 2023. p. 1762-77, doi: 10.18653/v1/2023.acl-long.99.

[14] Bordes A, Usunier N, Garcia-Duran A, Weston J, Yakhnenko O. Translating Embeddings for Modeling Multi-relational Data. In: Advances in Neural Information Processing Systems 26 (NeurIPS 2013); 2013. p. 2787-95.

[15] Sun Z, Deng Z-H, Nie J-Y, Tang J. RotatE: Knowledge Graph Embedding by Relational Rotation in Complex Space. In: Proceedings of the International Conference on Learning Representations (ICLR 2019); 2019, doi: 10.48550/arXiv.1902.10197.

[16] Gleich D. PageRank Beyond the Web. SIAM Rev. 2015;57(3):321-63, doi: 10.1137/140976649.

[17] Jeh G, Widom J. Scaling Personalized Web Search. In: Proceedings of the 12th International Conference on World Wide Web (WWW 2003); 2003 May 20-24; Budapest, Hungary. New York: ACM; 2003. p. 271-9, doi: 10.1145/775152.775191.

[18] Robertson S, Zaragoza H. The Probabilistic Relevance Framework: BM25 and Beyond. Found Trends Inf Retr. 2009;3(4):333-89, doi: 10.1561/1500000019.

[19] Manning C, Raghavan P, Schutze H. Introduction to Information Retrieval. New York: Cambridge University Press; 2008. 544 p., doi: 10.1017/CBO9780511809071.

[20] Chalkidis I, Fergadiotis M, Malakasiotis P, Aletras N, Androutsopoulos I. LEGAL-BERT: The Muppets Straight out of Law School. In: Findings of the Association for Computational Linguistics: EMNLP 2020; 2020. p. 2898-904, doi: 10.18653/v1/2020.findings-emnlp.261.

[21] Do N, Nguyen H, Selamat A. Knowledge-Based model of Expert Systems using Rela-model. Int J Softw Eng Knowl Eng. 2018;28(8):1047-90, doi: 10.1142/S0218194018500304.

[22] Nguyen DQ, Nguyen AT. PhoBERT: Pre-trained Language Models for Vietnamese. In: Findings of the Association for Computational Linguistics: EMNLP 2020; 2020. p. 1037-42, doi: 10.18653/v1/2020.findings-emnlp.92.

[23] Vu T, Nguyen DQ, Nguyen DQ, Dras M, Johnson M. VnCoreNLP: A Vietnamese Natural Language Processing Toolkit. In: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations (NAACL 2018); 2018. p. 56-60, doi:

10.18653/v1/N18-5012.

[24] Ngo H, Nguyen T, Nguyen D, Pham M. AimeLaw at ALQAC 2021: Enriching Neural Network Models with Legal-Domain Knowledge. In: Proceedings of the 13th International Conference on Knowledge and Systems Engineering (KSE 2021). IEEE; 2021, doi: 10.1109/KSE53942.2021.9648712.

[25] Devlin J, Chang M, Lee K, Toutanova K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2019); 2019. p. 4171-86, doi: 10.18653/v1/N19-1423.

[26] Zhong H, Xiao C, Tu C, Zhang T, Liu Z, Sun M. How Does NLP Benefit Legal System: A Summary of Legal Artificial Intelligence. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL 2020); 2020. p. 5218-30, doi: 10.18653/v1/2020.acl-main.466.

[27] Cui J, Li X, Shao Z, et al. A Survey on Legal Large Language Models. arXiv preprint. 2024. arXiv:2312.03718, doi: 10.48550/arXiv.2312.03718.

[28] Cormack GV, Clarke CLA, Buettcher S. Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods. In: Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2009); 2009. p. 758-9, doi: 10.1145/1571941.1572114.